Class's Test - 2

① $F_1(q, Q) = -qQ$

(1)

$$\{q_i, p_i, H\} \xrightarrow[C.T.]{} \{Q, P, K\}$$

$$\dot{q}_i = \frac{\partial H}{\partial p_i} \qquad \dot{Q}_i = \frac{\partial K}{\partial P_i}$$

$$\dot{p}_i = -\frac{\partial H}{\partial q_i} \qquad \dot{P}_i = -\frac{\partial K}{\partial Q_i}$$

If both (P, q) & (P, Q) are to satisfy Hamiltonian's EOM :

$$p_i \dot{q}_i - H = P_i \dot{Q}_i - K + \frac{\partial F_1}{\partial t} \frac{dF_1}{dt}$$

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$$\Rightarrow p_i \dot{q}_i - H = P_i \dot{Q}_i - K + \frac{\partial F_1}{\partial t} + \frac{\partial F_1}{\partial q_i} \dot{q}_i + \frac{\partial F_1}{\partial Q_i} \dot{Q}_i$$

$$= \dot{Q}_i \left(P_i + \frac{\partial F_1}{\partial Q_i} \right) + \left(\frac{\partial F_1}{\partial t} - K \right) + \frac{\partial F_1}{\partial q_i} \dot{q}_i$$

Comparing both sides:

$$p_i = \frac{\partial F_1}{\partial q_i} = -Q_i \Rightarrow p = -Q.$$

$$P_i = -\frac{\partial F_1}{\partial Q_i} = q_i \Rightarrow P = q$$

[Proved]

$$\frac{\partial F_1}{\partial t} + H = K.$$

(2) (ii) $K(Q, P) = H + \frac{\partial F_1}{\partial t} = H \quad \left[\frac{\partial F_1}{\partial t} = 0 \right].$

$$= \frac{p^2}{2m} + V(q).$$

$$\boxed{K(Q, P) = \frac{Q^2}{2m} + \phi(P)} \quad ; \quad [V(q) = \phi(P)]$$

$\therefore \quad Q = \frac{\partial K}{\partial P} = \frac{\partial \phi(P)}{\partial P} \quad ; \quad P = -\frac{\partial K}{\partial Q} = -\frac{Q}{m}. \quad$ — (I)

$\therefore \quad q = \frac{\partial H}{\partial p} = \frac{p}{m} \quad ; \quad p = -\frac{\partial H}{\partial q} = -\frac{\partial V(q)}{\partial q} \quad$ — (II)

If we swap $(P, Q) \longrightarrow (q, -p)$; ~~(I) ⟷ (II)~~

~~become~~:

(I) : $P = -\frac{Q}{m} \Rightarrow q = \frac{p}{m}$ —— (II) obtained

~~(II)~~ $\cancel{p \rightarrow}$ $Q = \frac{\partial \phi(P)}{\partial P} \Rightarrow -p = \frac{\partial V(q)}{\partial q}$ — " "

Hence, the transformation is verified.

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(3) $\pi_x = p_x + \frac{eB}{2} y \quad ; \quad \pi_y = p_y - \frac{eB}{2} x$

$x = x + \frac{\pi_y}{eB} \quad ; \quad y = y - \frac{\pi_x}{eB}$

(i) $[\pi_x, \pi_y] = [p_x + \frac{eB}{2} y, p_y - \frac{eB}{2} x]$.

$= \underbrace{[p_x, p_y]}_{\to 0} - \frac{eB}{2}[p_x, x] + \frac{eB}{2}[y, p_y] - \underbrace{\left(\frac{eB}{2}\right)^2 [y, x]}_{\to 0}$

$= \frac{eB}{2}\left([y, p_y] - [p_x, x]\right)$.

$[p_x, x] = \frac{\partial p_x}{\partial x}\frac{\partial x}{\partial p_x} - \frac{\partial p_x}{\partial p_x}\frac{\partial x}{\partial x} = -1. = [p_y, y]$. — (1).

substituting (1):

$[\pi_x, \pi_y] = \frac{eB}{2}(2) = eB$. (Ans)

(ii) $[x, y] = [x + \frac{\pi_y}{eB}, y - \frac{\pi_x}{eB}]$.

$= \underbrace{[x, y]}_{\to 0} - \frac{1}{eB}[x, \pi_x] + \frac{1}{eB}[\pi_y, y] - \frac{1}{(eB)^2}[\pi_y, \pi_x]$

$= \frac{1}{eB}\left([\pi_y, y] - [x, \pi_x]\right) + \frac{1}{eB}$.

$[\pi_y, y] = [p_y - \frac{eB}{2} x, y]$

$= [p_y, y] - \frac{eB}{2}\underbrace{[x, y]}_{\to 0} = -1$ — (II)

~~$= -[\pi_x$~~ (from (1))

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~~Substituting~~ (ii) : $[x, \pi_x] = [x, p_x + \frac{eB}{2} y]$

$\cancel{\frac{1}{eB} [x, y] = \frac{1}{eB}}$ $= [x, p_x] + \frac{eB}{2} \underbrace{[x, y]}_{\to 0}$

$= 1$ (from (i)) — (iii)

Using (ii) & (iii):

$[x, y] = \frac{1}{eB}(-1 - 1) + \frac{1}{eB} = -\frac{2}{eB} + \frac{1}{eB} = -\frac{1}{eB}$ (Ans)

$[x, y] \neq 0 \longrightarrow$ Guiding center coordinates do not Poisson commute.

(iii) $\cancel{[x, H]} = H = \frac{1}{2m}(\pi_x^2 + \pi_y^2)$

$\therefore [x, H] = \frac{1}{2m}[x, \pi_x^2 + \pi_y^2]$

$\cancel{= \frac{1}{2m}}$

$[x, \pi_x] = [x + \frac{\pi_y}{eB}, \pi_x]$.

$= [x, \pi_x] + \frac{1}{eB}[\pi_y, \pi_x]$

$= 1 + -1 = 0$. — (iv)

$[x, \pi_y] = [x + \frac{\pi_y}{eB}, \pi_y]$.

$= [x, \pi_y] + \frac{1}{eB} \underbrace{[\pi_y, \pi_y]}_{\to 0}$

$= [x, p_y - \frac{eB}{2} x]$.

$= [x, p_y] - \frac{eB}{2}[x, x] = 0$ — (v)

Using (iv) & (v): $[x, H] = \frac{1}{2m}([x, \pi_x^2] + [x, \pi_y^2])$

$= \frac{1}{2m}(\pi_x [x, \pi_x] + [x, \pi_x]\pi_x + \pi_y [x, \pi_y] + [x, \pi_y]\pi_y)$

$= 0$.

$\boxed{[x, H] = 0}$ (Ans) — (a)

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$$[Y, H] = \frac{1}{2m}[Y, \pi_x^2 + \pi_y^2].$$

$$\therefore [Y, \pi_x] = [y - \frac{\pi_x}{eB}, \pi_x]$$

$$= [y, \pi_x] - \frac{1}{eB}\cancel{[\pi_x, \pi_x]}^{\to 0}$$

$$= [y, p_x + \frac{eB}{2}y]$$

$$= \cancel{[y, p_x]}^{\to 0} + \frac{eB}{2}\cancel{[y, y]}^{\to 0} = 0 \quad \text{— (vi)}.$$

$$[Y, \pi_y] = [y - \frac{\pi_x}{eB}, \pi_y].$$

$$= [y, \pi_y] - \frac{1}{eB}[\pi_x, \pi_y]$$

$$= 1 - 1 = 0 \quad \text{— (vii)}.$$

$$[Y, H] = \frac{1}{2m}([Y, \pi_x^2] + [Y, \pi_y^2]).$$

$$= \frac{1}{2m}(\pi_x[Y, \pi_x] + [Y, \pi_x]\pi_x + \pi_y[Y, \pi_y] + [Y, \pi_y]\pi_y).$$

(b) — $\boxed{[Y, H] = 0}$ [from (vi) & (vii)].

Hamiltonian's EOM: $\frac{dA}{dt} = [A, H] + \frac{\partial A}{\partial t}$

For (X, Y) : $\frac{\partial X}{\partial t} = \frac{\partial Y}{\partial t} = 0$ (no dependence on time)

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$\therefore \frac{dX}{dt} = \frac{dY}{dt} = 0.$ [from ⓐ & ⓑ]

$\therefore$ X & Y are conserved, as phase-space f^ns.

② $[U, H] = \sum_i \left(\frac{\partial U}{\partial q_i} \frac{\partial H}{\partial p_i} - \frac{\partial U}{\partial p_i} \frac{\partial H}{\partial q_i} \right)$

From Hamiltonian's EOM:

$$\dot{p}_i = -\frac{\partial H}{\partial q_i} \quad ; \quad \dot{q}_i = \frac{\partial H}{\partial p_i}$$

$$\therefore [U, H] = \frac{\partial U}{\partial q_i} \dot{q}_i + \frac{\partial U}{\partial p_i} \dot{p}_i \quad \text{— ①}$$

$U = U(q_i, p_i, t)$

$$\Rightarrow \frac{dU}{dt} = \frac{\partial U}{\partial q_i} \dot{q}_i + \frac{\partial U}{\partial p_i} \dot{p}_i + \frac{\partial U}{\partial t}$$

$$= [U, H] + \frac{\partial U}{\partial t} \quad \text{(from ①)}$$

$$\Rightarrow \boxed{\frac{dU}{dt} - \frac{\partial U}{\partial t} = [U, H]} \quad \text{(Ans)}$$